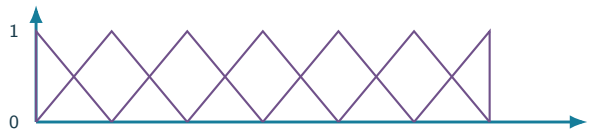
Mamdani gain scheduling: memberships, omissions and memory

04 / Exact sparse consequents, declared membership reconstruction and three independent outputs

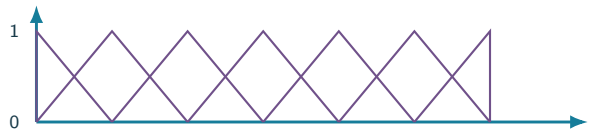
$\Delta e_k = e_k - e_{k-1}$; both inputs are clipped to their domains before fuzzification.

Error magnitude



$|e| \in [0, 40]\,^{\circ}\mathrm{C}$

Per-sample error change



$\Delta e \in [-40, 40]\,^{\circ}\mathrm{C}$

From activation to a normalized gain
Minimum antecedent strength; maximum consequent aggregation. Clip output triangles; average all maximizers on $[0, 1]$ (201 points by default): mean of maxima, not centroid.

Proportional output K'_p

	NB	NM	NS	Z	PS	PM	PB
NB	M	–	MS	M	MS	–	B
NM	MS	–	MS	M	MS	–	B
NS	S	–	MS	M	MS	–	B
Z	S	–	MS	M	MS	–	B
PS	S	–	MS	M	MS	–	B
PM	MS	–	MS	M	MS	–	B
PB	M	–	MS	M	MS	–	B

Columns: $|e|$ category; rows: signed Δe category

Integral output K'_i

	NB	NM	NS	Z	PS	PM	PB
NB	B	–	S	S	–	–	S
NM	MB	–	S	S	S	–	S
NS	M	–	S	S	S	–	S
Z	S	–	S	S	S	–	S
PS	S	–	S	S	S	–	S
PM	MB	–	S	S	S	–	S
PB	B	–	S	S	–	–	S

Columns: $|e|$ category; rows: signed Δe category

Derivative output K'_d

	NB	NM	NS	Z	PS	PM	PB
NB	S	–	MS	M	MB	–	B
NM	S	–	MS	MS	M	–	M
NS	S	–	S	S	MS	–	S
Z	S	–	S	S	S	–	S
PS	S	–	S	S	MS	–	S
PM	S	–	MS	MS	M	–	M
PB	S	–	MS	M	MB	–	B

Columns: $|e|$ category; rows: signed Δe category

Omitted does not mean zero
Gray NM/PM columns are unspecified in the source table. "–" marks an omitted gain. If a gain has no active consequent, retain its previous normalized value (initially 0.5). Each gain has separate memory.

Map each normalized output

$$K_j = K_{j,min} + (K_{j,max} - K_{j,min})K'_j.$$

$K_{min} = (.35, .0008, 0)$;
$K_{max} = (.90, .045, 2)$.

The output vector orders the parallel proportional, integral and derivative gains.

Reconstruction versus extension
Equal membership spacing is assumed. NB–PB label categories even on the nonnegative magnitude axis. Dense rules interpolate omissions; smoothing and IMC blending are separate research extensions. None is a recovered hardware setting.

The three matrices are generated directly from EXACT_SPARSE_RULES in fuzzy.py and retain the thesis's omissions. Membership sketches show the declared triangular geometry; the existing computed surface figures remain separate evidence.
Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.